

1. Banking Customer Support Chatbot

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PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/8.py"
1. TOKENS
['show', 'me', 'the', 'transactions', 'with', 'the', 'card', 'from', 'last', 'month']

2. CFG ANALYSIS
S -> VP
VP -> V NP
NP -> N PP PP
PP -> P NP

3. AMBIGUITY ANALYSIS
Possible interpretation 1:
Transactions made with the card during last month

Possible interpretation 2:
Transactions with the card, where 'from last month'
is attached as a separate time condition.

4. PCFG ANALYSIS
Parse 1 probability: 0.8
Parse 2 probability: 0.2
Selected Parse: Transactions made with the card during last month

5. FEATURE STRUCTURE
ACTION = SHOW
OBJECT = TRANSACTIONS
```

2. Voice Assistant Flight Booking

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PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/9.py"
2. AMBIGUITY ANALYSIS
Parse 1:
Book [a flight to Delhi] [with a window seat]

Parse 2:
Book [a flight [to Delhi with a window seat]]

Most likely meaning:
Destination = Delhi
Seat = Window

3. EARLEY PARSING
S -> VP .
VP -> V NP .
NP -> flight PP .
PP -> to NP .
PP -> with NP .

Earley parsing keeps multiple possible states
instead of restarting the complete parse.

4. PARTIAL INPUT
Input: Book a flight to
Parser state:
VP -> Book NP .
```

3. Healthcare NLP System

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PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)> & C:\Users\K.RAMESH\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/K.RAMESH/OneDrive/Desktop/CO4_AT1(Data Interpretation)/10.py"
)

RECOMMEND(
    agent = DOCTOR,
    action1 = START(MEDICATION),
    action2 = SCHEDULE(FOLLOW_UP_VISIT),
    location = CHENNAI
)

10. STRUCTURED OUTPUT
reviewer : Doctor
reviewed : Patient
review_time : Last Week
main_action : Recommend
treatment : Starting Medication
follow_up : Schedule Follow-up Visit
location : Chennai

FINAL RESULT
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Diagnosis : Not explicitly provided
Treatment : Starting medication
Follow-up : Schedule a follow-up visit
Location : Chennai
PS C:\Users\K.RAMESH\OneDrive\Desktop\CO4_AT1(Data Interpretation)>
```